

Outline

Setup: Local Units (§2.1)

Operators (§2.2)

Interpolating Power Series (§2.3)

Weierstrass Preparation Theorem

Cyclotomic extensions, units

Fix a prime number p . For each $n \geq 0$, define

$$K_n := \mathbb{Q}_p(\mu_{p^{n+1}}),$$

where $\mu_{p^{n+1}}$ is the group of p^{n+1} -th roots of unity.

We obtain:

$$K_0 \subset K_1 \subset K_2 \subset \cdots, \quad K_\infty := \bigcup_{n \geq 0} K_n.$$

Definitions

- ▶ $\mathcal{O}_{K_n}$: ring of integers of K_n
- ▶ $U_n := \mathcal{O}_{K_n}^\times$: unit group

Compatible roots of unity

- ▶ Each U_n is a multiplicative group
- ▶ These are the **local units** at level n

Fix $\zeta_n \in \mu_{p^{n+1}}$ such that

$$\zeta_{n+1}^p = \zeta_n \quad \text{for all } n \geq 0.$$

Define

$$\pi_n := \zeta_n - 1.$$

It generates the maximal ideal of $\mathfrak{m}_{K_n} \subseteq \mathcal{O}_{K_n}$, i.e. $\mathfrak{m}_{K_n} = (\pi_n)$:

Since ζ_n is a root of the cyclotomic polynomial

$$\Phi_{p^{n+1}}(X) = \frac{X^{p^{n+1}} - 1}{X^{p^n} - 1} = 1 + X^{p^n} + X^{2p^n} + \dots + X^{(p-1)p^n}.$$

the element π_n is a root of

$$\Phi_{p^{n+1}}(1 + T) \in \mathbb{Z}_p[T].$$

Now $\Phi_{p^{n+1}}(1 + T)$ is an Eisenstein polynomial at p :

- ▶ it is monic,
- ▶ its constant term is p (divisible by p , but not by p^2).
- ▶ all its other non-leading coefficients are divisible by p .

Therefore $\Phi_{p^{n+1}}(1 + T)$ is irreducible over $\mathbb{Q}_p$, $K_n = \mathbb{Q}_p(\pi_n)$ (totally ramified), and its root π_n generates the maximal ideal of $\mathcal{O}_{K_n}$.

Power series ring

Write the formal power series in one variable $R := \mathbb{Z}_p[[T]]$.

Key observation

Recall that Every element $x \in \mathcal{O}_{K_n}$ can be written as a power series

$$x = \sum_{k \geq 0} a_k \pi_n^k, \quad a_k \in \mathbb{Z}_p,$$

So as $z \in U_n$. Define $f(T) := \sum_{k \geq 0} a_k T^k \in \mathbb{Z}_p[[T]] = R$, then $f(\pi_n) = z$.

Why is $f(T)$ not unique?

The representation of z as a power series in π_n is *not unique*: the map

$$\mathbb{Z}_p[[T]] \longrightarrow \mathcal{O}_{K_n}, \quad f(T) \mapsto f(\pi_n)$$

has a *nontrivial kernel*; Indeed, π_n satisfies the relation

$$\Phi_{p^{n+1}}(1 + \pi_n) = 0.$$

So if $g(T)$ is any power series divisible by $\Phi_{p^{n+1}}(1 + T)$, then $g(\pi_n) = 0$. Hence if $f(T)$ satisfies $f(\pi_n) = z$, then so does

$$f(T) + g(T),$$

for any such $g(T) \neq 0$.

Idea to fix this

Instead of prescribing the value at a single point, we impose

$$f(\pi_n) = u_n \quad \text{for all } n \geq 0,$$

where $(u_n) \in U_\infty$ is a norm-compatible system.

Norm maps, compatibility

Let K_n/K_m be a finite extension with $n \geq m$. The norm map

$$N_{n,m} : K_n^\times \rightarrow K_m^\times, \quad x \mapsto \prod_{\sigma \in \text{Gal}(K_n/K_m)} \sigma(x).$$

It is multiplicative: $N_{n,m}(xy) = N_{n,m}(x)N_{n,m}(y)$.

In our setting, $K_n = \mathbb{Q}_p(\mu_{p^{n+1}})$, $K_{n-1} = \mathbb{Q}_p(\mu_{p^n})$.

- ▶ The extension K_n/K_{n-1} has Galois group μ_p
- ▶ The action is:

$$\sigma_\xi(\zeta_n) = \xi \zeta_n, \quad \xi \in \mu_p$$

- ▶ $N_{n,n-1}(\zeta_n) = \prod_{\xi \in \mu_p} \xi \zeta_n = \left(\prod_{\xi \in \mu_p} \xi \right) \zeta_n^p = \zeta_n^p = \zeta_{n-1}$.

Norm maps, compatibility

Define U_∞ as the projective limit taken with respect to the norm maps:

$$U_\infty = \varprojlim U_n, \quad \text{i.e.}$$

A sequence (u_n) lies in U_∞ if

$$u_n \in U_n, \quad N_{n,m}(u_n) = u_m$$

name it *norm-compatible system of units*.

We have seen that the fixed sequence of local units (ζ_n) forms a norm compatible systems of units.

Example (1)

Assume $p \neq 2$ ($2 \in \mathbb{Z}_p^\times$). Let $a, b \in \mathbb{Z}$ be non-zero integers with $(a, p) = (b, p) = 1$. Consider the sequence

$$u_n = \frac{\zeta_n^{-a/2} - \zeta_n^{a/2}}{\zeta_n^{-b/2} - \zeta_n^{b/2}} = \zeta_n^{(b-a)/2} \cdot \frac{1 - \zeta_n^a}{1 - \zeta_n^b}.$$

We show that (u_n) is a *norm-compatible system of units*.

First we show that $u_n \in \mathcal{O}_{K_n}^\times$. It suffices to show

$$\frac{1 - \zeta_n^a}{1 - \zeta_n^b} \in \mathcal{O}_{K_n}^\times$$

Since $1 - \zeta_n^k$ is a Galois conjugate of $1 - \zeta_n$, we have for $(k, p) = 1$,

$$v_{K_n}(1 - \zeta_n^k) = v_{K_n}(1 - \zeta_n).$$

Example (2)

We first prove that for any integer k with $(k, p) = 1$,

$$N_{n,n-1}(1 - \zeta_n^k) = 1 - \zeta_{n-1}^k.$$

Indeed, since K_n/K_{n-1} is Galois with group μ_p , we have

$$N_{n,n-1}(1 - \zeta_n^k) = \prod_{\xi \in \mu_p} (1 - (\xi \zeta_n)^k).$$

Because $(k, p) = 1$, the map $\xi \mapsto \xi^k$ permutes μ_p , so this equals

$$\prod_{\eta \in \mu_p} (1 - \eta \zeta_n^k).$$

Using

$$\prod_{\eta \in \mu_p} (X - \eta) = X^p - 1$$

with $X = \zeta_n^k$, we obtain

$$N_{n,n-1}(1 - \zeta_n^k) = 1 - (\zeta_n^k)^p = 1 - \zeta_{n-1}^k.$$

Example (3)

Write

$$\zeta_n^{-a/2} - \zeta_n^{a/2} = \zeta_n^{-a/2}(1 - \zeta_n^a).$$

Then

$$N_{n,n-1}(\zeta_n^{-a/2} - \zeta_n^{a/2}) = N_{n,n-1}(\zeta_n^{-a/2}) N_{n,n-1}(1 - \zeta_n^a).$$

Now

$$N_{n,n-1}(\zeta_n) = \zeta_{n-1},$$

so

$$N_{n,n-1}(\zeta_n^{-a/2}) = \zeta_{n-1}^{-a/2}.$$

By the previous frame,

$$N_{n,n-1}(1 - \zeta_n^a) = 1 - \zeta_{n-1}^a.$$

Hence

$$N_{n,n-1}(\zeta_n^{-a/2} - \zeta_n^{a/2}) = \zeta_{n-1}^{-a/2}(1 - \zeta_{n-1}^a) = \zeta_{n-1}^{-a/2} - \zeta_{n-1}^{a/2}.$$

Example (4)

The same argument gives

$$N_{n,n-1}(\zeta_n^{-b/2} - \zeta_n^{b/2}) = \zeta_{n-1}^{-b/2} - \zeta_{n-1}^{b/2}.$$

Therefore, by multiplicativity of the norm,

$$N_{n,n-1}(u_n) = \frac{N_{n,n-1}(\zeta_n^{-a/2} - \zeta_n^{a/2})}{N_{n,n-1}(\zeta_n^{-b/2} - \zeta_n^{b/2})} = \frac{\zeta_{n-1}^{-a/2} - \zeta_{n-1}^{a/2}}{\zeta_{n-1}^{-b/2} - \zeta_{n-1}^{b/2}} = u_{n-1}.$$

So the units (u_n) are norm compatible.

Main Theorem Today

Theorem (Theorem 2.1.2)

For each $u = (u_n) \in U_\infty$, there exists a unique

$$f_u(T) \in R$$

such that

$$f_u(\pi_n) = u_n \quad \text{for all } n \geq 0.$$

- ▶ $f_u(T)$ is called the **Coleman power series**
- ▶ A single power series encodes the entire tower
- ▶ Across all n , compatibility imposes strong rigidity

Trivial example: $f_{(\zeta_n)}(T) = 1 + T$.

Example (5): $w_k(T) \in \mathbb{Z}_p[[T]]^\times$

For any integer k with $(k, p) = 1$, define

$$w_k(T) := \frac{(1+T)^{-k/2} - (1+T)^{k/2}}{T}.$$

Here $(1+T)^{\pm k/2}$ is interpreted by the binomial expansion in $\mathbb{Z}_p[[T]]$; this makes sense because $2 \in \mathbb{Z}_p^\times$.

Also notice that $(1+T)^{-k/2} - (1+T)^{k/2}$ has zero constant term. Hence it is divisible by T in $\mathbb{Z}_p[[T]]$. $w_k(T) \in \mathbb{Z}_p[[T]]$.

Expand to first order:

$$(1+T)^{\pm k/2} = 1 \pm \frac{k}{2}T + \dots$$

so

$$(1+T)^{-k/2} - (1+T)^{k/2} = -kT + \dots$$

Dividing by T gives $w_k(0) = -k$. Since $(k, p) = 1$, we have $-k \in \mathbb{Z}_p^\times$, hence $w_k(T) \in \mathbb{Z}_p[[T]]^\times$.

Example (6): Evaluation at π_n

Recall that $1 + \pi_n = \zeta_n$. Substituting $T = \pi_n$ into the definition of w_k gives

$$w_k(\pi_n) = \frac{\zeta_n^{-k/2} - \zeta_n^{k/2}}{\zeta_n - 1}.$$

Thus we define $f_u(T) := \frac{w_a(T)}{w_b(T)} \in R^\times$. Then

$$f_u(\pi_n) = \frac{w_a(\pi_n)}{w_b(\pi_n)} = \frac{\frac{\zeta_n^{-a/2} - \zeta_n^{a/2}}{\zeta_n - 1}}{\frac{\zeta_n^{-b/2} - \zeta_n^{b/2}}{\zeta_n - 1}} = \frac{\zeta_n^{-a/2} - \zeta_n^{a/2}}{\zeta_n^{-b/2} - \zeta_n^{b/2}} = u_n.$$

We have shown that the explicit power series $f_u(T) = \frac{w_a(T)}{w_b(T)}$ satisfies

$$f_u(\pi_n) = u_n \quad \text{for all } n \geq 0.$$

So this example is a concrete instance of the interpolation phenomenon in Theorem.

Example: cyclotomic units

let $a \in \mathbb{Z}$ prime to p , and define

$$c_n(a) := \frac{\zeta_{p^n}^a - 1}{\zeta_{p^{n-1}}^a - 1} \in U_n.$$

Lemma

We have $c(a) := (c_n(a))_n \in U_\infty$.

Proof. recall that we computed:

$$N_{n,n-1}(\zeta_{p^n}^a - 1) = \prod_{\eta \in \mu_p} (\eta \zeta_{p^n}^a - 1) = \zeta_{p^n}^{ap} - 1 = \zeta_{p^{n-1}}^a - 1.$$

Hence

$$N_{n,n-1}(c_n(a)) = \frac{\zeta_{p^{n-1}}^a - 1}{\zeta_{p^{n-1}}^a - 1} = c_{n-1}(a).$$

Directly by observation, we get that the function

$$f_{c(a)}(T) = \frac{(1+T)^a - 1}{T}$$

satisfies the required property.

Uniqueness: Weierstrass preparation theorem (1)

Theorem 2.1.2 says that if $(u_n) \in U_\infty$, then there is a *unique* formal power series

$$f_u(T) \in R = \mathbb{Z}_p[[T]]$$

such that

$$f_u(\pi_n) = u_n \quad \text{for all } n \geq 0.$$

To prove uniqueness, suppose $f(T), g(T) \in R$ both satisfy

$$f(\pi_n) = g(\pi_n) \quad \text{for all } n \geq 0.$$

Set

$$h(T) = f(T) - g(T).$$

Then

$$h(\pi_n) = 0 \quad \text{for all } n \geq 0.$$

So $h(T)$ has infinitely many zeros in the maximal ideal, namely the points $\pi_n = \zeta_n - 1$.

Uniqueness: Weierstrass preparation theorem (2)

Theorem (Weierstrass preparation theorem)

Every power series $f(T) \in R = \mathbb{Z}_p[[T]]$ can be written uniquely in the form

$$f(T) = p^m g(T) w(T),$$

where:

- ▶ $m \geq 0$,
- ▶ $g(T)$ is a distinguished polynomial, i.e. monic and all lower coefficients lie in $p\mathbb{Z}_p$,
- ▶ $w(T) \in R^\times$ is a unit.

this means: Apply to $x \in \mathfrak{m}$, the power series converges. Thus the zero set of a nonzero $f(T)$ in $\mathfrak{m}$ is controlled by the polynomial factor $g(T)$, which must be finite.

Operators we will use

In this section we work on the commutative algebra of

$$R = \mathbb{Z}_p[[T]].$$

There are three main operators on R :

► **Frobenius:**

$$\varphi(f)(T) = f((1+T)^p - 1).$$

► **Norm operator:**

$$N : R \rightarrow R.$$

► **Trace operator:**

$$\psi : R \rightarrow R.$$

The operators N and ψ are defined so that their images under φ are explicit averages over μ_p .

We will prove the following facts.

1. φ is injective.
2. There exist unique continuous maps $N, \psi : R \rightarrow R$ such that

$$(\varphi \circ N)(f)(T) = \prod_{\xi \in \mu_p} f(\xi(1+T) - 1),$$

$$(\varphi \circ \psi)(f)(T) = \frac{1}{p} \sum_{\xi \in \mu_p} f(\xi(1+T) - 1).$$

3. ψ is $\mathbb{Z}_p$ -linear and satisfies

$$\psi \circ \varphi = \text{id}_R.$$

4. N is multiplicative:

$$N(fg) = N(f)N(g).$$

5. Hence $N(R^\times) \subseteq R^\times$.

Two structural lemmas

The construction of N and ψ depends on two key observations.

- ▶ The image of φ consists exactly of those power series $h(T) \in R$ satisfying

$$h(\xi(1+T) - 1) = h(T) \quad (\xi \in \mu_p).$$

- ▶ If a power series is invariant under all substitutions

$$T \mapsto \xi(1+T) - 1,$$

then it must come from φ .

We prove the previous page in three steps:

1. first show that φ is injective;
2. characterize the image of φ ;
3. use that characterization to define N and ψ (uniquely).

The key point is that all formulas we write down are invariant under the substitutions

$$T \mapsto \xi(1+T) - 1 \quad (\xi \in \mu_p).$$

Injectivity of φ

Recall that

$$\varphi(f)(T) = f((1+T)^p - 1).$$

If

$$f(T) = a_m T^m + a_{m+1} T^{m+1} + \cdots \quad \text{with } a_m \neq 0,$$

then

$$\varphi(f)(T) = a_m ((1+T)^p - 1)^m + \cdots .$$

Since

$$(1+T)^p - 1 = pT + \cdots ,$$

Then the coefficient of T^m is $a_m p^m$, which is clearly non-zero.

Hence φ is injective.

The image of φ

Let

$$\omega(T) := (1 + T)^p - 1.$$

Then

$$\varphi(R) = \mathbb{Z}_p[[\omega(T)]].$$

We claim that $\varphi(R)$ consists of all power series $h(T) \in R$ such that

$$h(\xi(1 + T) - 1) = h(T) \quad (\forall \xi \in \mu_p).$$

Every element of $\varphi(R)$ is invariant

If $h(T) = \varphi(g)(T) = g(\omega(T))$, then

$$h(\xi(1+T) - 1) = g((1 + \xi(1+T) - 1)^p - 1).$$

Since $\xi^p = 1$, this becomes

$$g((1+T)^p - 1) = h(T).$$

So every element of $\varphi(R)$ is invariant under

$$T \mapsto \xi(1+T) - 1.$$

Why every invariant series lies in $\varphi(R)$ (1)

Now suppose $h(T) \in R$ satisfies

$$h(\xi(1+T) - 1) = h(T) \quad (\xi \in \mu_p).$$

Then in particular,

$$h(\xi - 1) = h(0) \quad (\xi \in \mu_p).$$

So the power series $h(T) - h(0)$ vanishes at all roots of

$$\omega(T) = (1+T)^p - 1.$$

Since $\omega(T)$ is distinguished, Weierstrass division gives

$$h(T) - h(0) = \omega(T) h_1(T)$$

for some $h_1(T) \in R$.

Why every invariant series lies in $\varphi(R)$ (2)

Because $\omega(\xi(1+T) - 1) = \omega(T)$, the quotient $h_1(T)$ is again invariant:

$$h_1(\xi(1+T) - 1) = h_1(T) \quad (\xi \in \mu_p).$$

Therefore the same argument applies again, and we get

$$h(T) = a_0 + a_1\omega(T) + \cdots + a_{n-1}\omega(T)^{n-1} + \omega(T)^n h_n(T)$$

with $a_i \in \mathbb{Z}_p$ and $h_n(T) \in R$.

By induction, we have

$$h(T) \in \mathbb{Z}_p[[\omega(T)]] = \varphi(R).$$

So the image characterization is proved.

Defining the norm operator N

For $f \in R$, set

$$H_f(T) := \prod_{\xi \in \mu_p} f(\xi(1+T) - 1).$$

Then $H_f(T)$ is invariant under

$$T \mapsto \eta(1+T) - 1 \quad (\eta \in \mu_p),$$

because that substitution merely permutes the factors in the product. By the image characterization of φ , there is a unique element $N(f) \in R$ such that

$$\varphi(N(f))(T) = H_f(T).$$

This defines N uniquely.

Defining the trace operator ψ

For $f \in R$, set

$$R_f(T) := \sum_{\xi \in \mu_p} f(\xi(1+T) - 1).$$

Again $R_f(T)$ is invariant under

$$T \mapsto \eta(1+T) - 1.$$

Let $\mathcal{O}_0$ be the ring of integers of $\mathbb{Q}_p(\mu_p)$, and let $\mathfrak{p}_0$ be its maximal ideal. For every $\xi \in \mu_p$, $\xi \equiv 1 \pmod{\mathfrak{p}_0}$, Hence:

$$\xi(1+T) - 1 \equiv T \pmod{\mathfrak{p}_0 R}, \quad f(\xi(1+T) - 1) \equiv f(T) \pmod{\mathfrak{p}_0 R}.$$

Summing over $\xi \in \mu_p$ gives:

$$R_f(T) = \sum_{\xi \in \mu_p} f(\xi(1+T) - 1) \equiv \sum_{\xi \in \mu_p} f(T) = p f(T) \equiv 0 \pmod{\mathfrak{p}_0 R}.$$

Therefore there is a unique $\psi(f) \in R$ such that

$$\varphi(\psi(f))(T) = \frac{1}{p} R_f(T).$$

$$\psi \circ \varphi = \text{id}_R$$

By construction, ψ is $\mathbb{Z}_p$ -linear. Also,

$$\varphi(\psi(\varphi(g)))(T) = \frac{1}{p} \sum_{\xi \in \mu_p} \varphi(g)(\xi(1+T) - 1).$$

But

$$\varphi(g)(\xi(1+T)-1) = g((\xi(1+T))^p - 1) = g((1+T)^p - 1) = \varphi(g)(T),$$

$$\varphi(\psi(\varphi(g))) = \varphi(g).$$

Since φ is injective,

$$\psi \circ \varphi = \text{id}_R.$$

Multiplicativity of N

Using the definition of N ,

$$\varphi(N(fg))(T) = \prod_{\xi \in \mu_p} (fg)(\xi(1+T) - 1).$$

But this is

$$\left(\prod_{\xi \in \mu_p} f(\xi(1+T) - 1) \right) \left(\prod_{\xi \in \mu_p} g(\xi(1+T) - 1) \right).$$

Therefore

$$\varphi(N(fg)) = \varphi(N(f)) \varphi(N(g)).$$

Since φ is injective,

$$N(fg) = N(f) N(g).$$

So N preserves products, and in particular

$$N(R^\times) \subset R^\times.$$

Computation on N and ψ (1)

From the definition of N ,

$$\varphi(N(T))(T) = \prod_{\xi \in \mu_p} (\xi(1+T) - 1) = (1+T)^p - 1 = \varphi(T).$$

Since φ is injective, $N(T) = T$.

For ψ , we prove that for all $n \geq 1$,

$$\psi\left(\varphi(T)^n \cdot \frac{1+T}{T}\right) = T^n \cdot \frac{1+T}{T}.$$

Apply the definition ψ : $\varphi(\psi(f))(T) = \frac{1}{p} \sum_{\xi \in \mu_p} f(\xi(1+T) - 1)$ to

$$f(T) = \varphi(T)^n \cdot \frac{1+T}{T},$$

We compute each term:

$$f(\xi(1+T) - 1) = \varphi(\xi(1+T) - 1)^n \cdot \frac{\xi(1+T)}{\xi(1+T) - 1}.$$

$$\varphi(\xi(1+T) - 1) = (1 + \xi(1+T) - 1)^p - 1 = (\xi(1+T))^p - 1.$$

Since $\xi^p = 1$, this becomes $(1+T)^p - 1 = \varphi(T)$.

Computation on ψ (2, prepared for next talk)

So:

$$\varphi(\xi(1+T) - 1)^n = \varphi(T)^n.$$

Thus

$$f(\xi(1+T) - 1) = \varphi(T)^n \cdot \frac{\xi(1+T)}{\xi(1+T) - 1}.$$

Hence

$$\varphi(\psi(f))(T) = \frac{1}{p} \sum_{\xi \in \mu_p} \varphi(T)^n \cdot \frac{\xi(1+T)}{\xi(1+T) - 1} = \varphi(T)^n \cdot \frac{1}{p} \sum_{\xi \in \mu_p} \frac{\xi(1+T)}{\xi(1+T) - 1}$$

We claim:

$$\frac{1}{p} \sum_{\xi \in \mu_p} \frac{\xi(1+T)}{\xi(1+T) - 1} = \frac{(1+T)^p}{(1+T)^p - 1} = \varphi\left(\frac{1+T}{T}\right).$$

Computation on ψ (3, prepared for next talk)

Indeed, set $X = 1 + T$. Using the identity $\prod_{\xi \in \mu_p} (X - \xi) = X^p - 1$, and taking logarithmic derivative:

$$\frac{d}{dX} \log \prod_{\xi} (X - \xi) = \sum_{\xi \in \mu_p} \frac{1}{X - \xi}.$$

On the other hand:

$$\frac{d}{dX} \log(X^p - 1) = \frac{pX^{p-1}}{X^p - 1}.$$

Therefore,

$$\sum_{\xi \in \mu_p} \frac{1}{X - \xi} = \frac{pX^{p-1}}{X^p - 1}.$$

Computation on ψ (4, prepared for next talk)

Since

$$\sum_{\xi \in \mu_p} \frac{\xi X}{\xi X - 1} = X \sum_{\xi \in \mu_p} \frac{1}{X - \xi}.$$

we get:

$$\frac{1}{p} \sum_{\xi \in \mu_p} \frac{\xi X}{\xi X - 1} = \frac{X^p}{X^p - 1}.$$

Substituting $X = 1 + T$ gives

$$\frac{1}{p} \sum_{\xi \in \mu_p} \frac{\xi(1+T)}{\xi(1+T) - 1} = \frac{(1+T)^p}{(1+T)^p - 1}.$$

Putting everything together:

$$\varphi(\psi(f))(T) = \varphi(T)^n \cdot \frac{(1+T)^p}{(1+T)^p - 1} = \varphi\left(T^n \cdot \frac{1+T}{T}\right).$$

Since φ is injective, we conclude:

$$\psi(\varphi(T)^n \cdot \frac{1+T}{T}) = \psi(f) = T^n \cdot \frac{1+T}{T}.$$

To prove the main theorem

At this point, we have introduced:

- ▶ the cyclotomic tower K_n and the inverse limit U_∞ ,
- ▶ the power series ring $R = \mathbb{Z}_p[[T]]$,
- ▶ the operators φ , N , and ψ .

To prove the main theorem, we will do three things:

1. prove the key congruence lemmas for N ;
2. show that the iterates $N^k(f)$ converge to a fixed point of N ;
3. use that fixed point construction to build $f_u(T)$ for an arbitrary $u \in U_\infty$.

Why $N(f) = f$ gives norm-compatible units (1)

Let $f \in R^\times$ and define

$$u_n := f(\pi_n), \quad \pi_n = \zeta_n - 1.$$

Claim: If $N(f) = f$, then $(u_n) \in U_\infty$, i.e.

$$N_{n,n-1}(u_n) = u_{n-1}.$$

Proof: Since $N_{n,n-1}(f(\zeta_n - 1)) = \prod_{\xi \in \mu_p} f(\xi \zeta_n - 1)$, input $\pi_n = \zeta_n - 1$, this becomes

$$N_{n,n-1}(u_n) = \prod_{\xi \in \mu_p} f(\xi(1 + \pi_n) - 1).$$

Why $N(f) = f$ gives norm-compatible units (2)

By definition of the norm operator and $N(f) = f$,

$$\varphi(f)(T) = \varphi(N(f))(T) = \prod_{\xi \in \mu_p} f(\xi(1 + T) - 1).$$

Thus, evaluating at $T = \pi_n$ gives

$$\varphi(f)(\pi_n) = \prod_{\xi \in \mu_p} f(\xi(1 + \pi_n) - 1) = N_{n,n-1}(u_n).$$

By definition of φ , $\varphi(f)(\pi_n) = f(\pi_{n-1})$, therefore:

$$N_{n,n-1}(u_n) = f(\pi_{n-1}) = u_{n-1}.$$

Lemmas on the norm N

The next lemmas will show:

- ▶ if $\varphi(f) \equiv 1 \pmod{p^k}$, then $f \equiv 1 \pmod{p^k}$;
- ▶ if $f \in R^\times$, then $N(f) \equiv f \pmod{p}$;
- ▶ if $f \equiv 1 \pmod{p^m}$, then $N(f) \equiv 1 \pmod{p^{m+1}}$;
- ▶ Assume $f \in R^\times$, and let $k_2 \geq k_1 \geq 0$, then $N^{k_2}(f) \equiv N^{k_1}(f) \pmod{p^{k_1+1}R}$. It is a direct corollary by noticing $N^{k_2-k_1}(f) \equiv f \pmod{pR}$, and applying N^{k_1} to both sides.
- ▶ hence the limit of $N^k(f)$ exists and is a fixed point of N .

Lemma 1

Assume $f \in R = \mathbb{Z}_p[[T]]$ and $k \geq 0$. Then

$$[\varphi(f)(T) \equiv 1 \pmod{p^k R}] \implies [f(T) \equiv 1 \pmod{p^k R}].$$

Proof. Write $f(T) = 1 + p^m h(T)$, where $m \geq 0$ is maximal and $h(T) \in R$ is not divisible by p . We have

$$\varphi(f)(T) = f((1+T)^p - 1) = 1 + p^m h((1+T)^p - 1).$$

Notice that $(1+T)^p - 1 \equiv T^p \pmod{pR}$, we get

$$h(\omega(T)) \equiv h(T^p) \pmod{pR}.$$

which is non-zero (by assumption on h). Therefore

$$\varphi(f)(T) - 1 \not\equiv 0 \pmod{p^{m+1}R}.$$

If $m < k$, this contradicts the hypothesis.

Proof of $N(f) \equiv f \pmod{pR}$

For each $\xi \in \mu_p$, we have

$$\xi(1 + T) - 1 \equiv T \pmod{\mathfrak{p}_0 R}.$$

Hence

$$f(\xi(1 + T) - 1) \equiv f(T) \pmod{\mathfrak{p}_0 R}.$$

Therefore

$$\varphi(N(f))(T) \equiv f(T)^p \pmod{pR}.$$

But in characteristic p ,

$$f(T)^p \equiv \varphi(f)(T) \pmod{pR}.$$

So

$$\varphi(N(f)) \equiv \varphi(f) \pmod{pR}.$$

By Lemma 1, we conclude

$$N(f) \equiv f \pmod{pR}.$$

Proof of $[f \equiv 1 \pmod{p^m R}] \implies [N(f) \equiv 1 \pmod{p^{m+1} R}]$

Assume that $f \equiv 1 \pmod{p^m R}$, $m \geq 1$, write $f(T) = \sum_{n \geq 0} a_n T^n$ with

$$a_0 \equiv 1 \pmod{p^m}, \quad a_n \equiv 0 \pmod{p^m} \text{ for } n \geq 1.$$

Same calculation gives

$$f(\xi(1+T) - 1) \equiv f(T) \pmod{p^m R}.$$

Therefore

$$\varphi(N(f))(T) = \prod_{\xi \in \mu_p} f(\xi(1+T) - 1) \equiv f(T)^p \equiv 1 \pmod{p^{m+1} R}.$$

By Lemma 1, this implies

$$N(f) \equiv 1 \pmod{p^{m+1} R}.$$

Convergence gives $N(g) = g$

As we mentioned before, with these three lemmas, assume $f \in R^\times$, and let $k_2 \geq k_1 \geq 0$, then

$$N^{k_2}(f) \equiv N^{k_1}(f) \pmod{p^{k_1+1}R}.$$

If $f \in R^\times$, then the sequence $N^k(f)$ converges in $R^\times$:
Since the sequence $\{N^k(f)\}$ is Cauchy for the p -adic topology, and R is complete with respect to p -adic topology, there exists

$$g = \lim_{k \rightarrow \infty} N^k(f) \in R^\times.$$

Continuity of N gives

$$N(g) = g.$$

Preparing the interpolation argument

Now let

$$u = (u_n) \in U_\infty.$$

For each n , choose $f_n \in R^\times$ such that

$$f_n(\pi_n) = u_n.$$

Define

$$g_n(T) := N^n(f_{2n})(T).$$

We will show that for each fixed n ,

$$g_m(\pi_n) \rightarrow u_n \quad (m \rightarrow \infty).$$

Lemma on the convergence

For all $n \geq 0$ and all $m \geq n$, we have

$$g_m(\pi_n) \equiv u_n \pmod{p^{m+1}}.$$

In particular,

$$\lim_{m \rightarrow \infty} g_m(\pi_n) = u_n.$$

Proof. Since $u = (u_n)$ is norm-compatible,

$$u_n = N_{2m,n}(u_{2m}).$$

Because $f_{2m}(\pi_{2m}) = u_{2m}$, this gives

$$u_n = (N^{2m-n} f_{2m})(\pi_n).$$

Now Corollary 2.3.3 implies

$$N^{2m-n}(f_{2m}) \equiv N^m(f_{2m}) = g_m \pmod{p^{m+1}R}.$$

Evaluating at π_n gives the claim.

Proof of the existence

The ring $R = \mathbb{Z}_p[[T]]$ is complete for the (p, T) -adic topology, so the sequence (g_n) has a limit point; call it $h(T)$.

For each fixed n , previous lemma on convergence shows that

$$g_m(\pi_n) \rightarrow u_n.$$

Passing to the limit along a convergent subsequence, we obtain

$$h(\pi_n) = u_n \quad \text{for all } n \geq 0.$$

Thus $h(T)$ is the desired interpolating power series.

Uniqueness follows from Weierstrass preparation theorem.

The Galois action and the fixed-point space

Let $G = \text{Gal}(K_\infty/\mathbb{Q}_p)$ and let

$$\chi : G \rightarrow \mathbb{Z}_p^\times$$

be the cyclotomic character (recall that it is defined by $\sigma(\zeta) = \zeta^{\chi(\sigma)}$ for all $\sigma \in G$ and $\zeta \in \mu_{p^\infty}$.)

For $\sigma \in G$, define

$$(\sigma f)(T) = f((1+T)^{\chi(\sigma)} - 1).$$

One can check that this action preserves $R^\times$, commutes with φ , N , and ψ , and gives a natural action on the fixed-point space

$$W = \{f \in R^\times : N(f) = f\}.$$

Corollary

There is an isomorphism of G -modules

$$U_\infty \xrightarrow{\sim} W, \quad u \longmapsto f_u.$$

Proof: G -equivariance

Let $\sigma \in G$ and $f \in W$. We compute:

$$(\sigma f)(\pi_n) = f((1 + \pi_n)^{\chi(\sigma)} - 1).$$

Since $\pi_n = \zeta_n - 1$, we have

$$(1 + \pi_n)^{\chi(\sigma)} - 1 = \zeta_n^{\chi(\sigma)} - 1 = \sigma(\zeta_n) - 1.$$

Hence

$$(\sigma f)(\pi_n) = f(\sigma(\zeta_n) - 1) = \sigma(f(\zeta_n - 1)).$$

Thus

$$(\sigma f)(\pi_n) = \sigma(f(\pi_n)).$$

If $f = f_u$, then $f(\pi_n) = u_n$, so

$$(\sigma f_u)(\pi_n) = \sigma(u_n).$$

Weierstrass preparation theorem

Theorem (2.1.3)

Each $f(T) \in R = \mathbb{Z}_p[[T]]$ can be written uniquely in the form

$$f(T) = p^m g(T) w(T),$$

where:

- ▶ $m \geq 0$,
- ▶ $g(T)$ is a distinguished polynomial, i.e.

$$g(T) = T^d + a_{d-1} T^{d-1} + \cdots + a_0, \quad a_i \in p\mathbb{Z}_p,$$

- ▶ $w(T) \in R^\times$ is a unit.

Step 1: reduce to a p -primitive series

Let

$$f(T) = \sum_{n \geq 0} a_n T^n \neq 0.$$

Choose

$$m = \min_n v_p(a_n).$$

Then we may write

$$f(T) = p^m F(T),$$

where $F(T) \in R$ and at least one coefficient of F is a p -adic unit.

So it remains to prove the theorem for a p -primitive series $F(T)$, i.e. one not divisible by p .

Step 2: the first nonzero term mod p

Reduce $F(T)$ modulo p :

$$\overline{F}(T) \in \mathbb{F}_p[[T]].$$

Since F is not divisible by p , the series $\overline{F}(T)$ is nonzero. Hence there is a unique integer $d \geq 0$ such that

$$\overline{F}(T) = T^d \overline{U}(T), \quad \overline{U}(0) \neq 0.$$

Equivalently:

- ▶ the first nonzero coefficient of $\overline{F}$ occurs in degree d ,
- ▶ and that coefficient is nonzero in $\mathbb{F}_p$.

This integer d will be the degree of the distinguished polynomial.

Step 3: the division lemma

We use the standard Weierstrass division theorem:

Weierstrass division

If $g(T) \in R$ is distinguished of degree d , then for every $h(T) \in R$ there exist unique

$$q(T) \in R, \quad r(T) \in \mathbb{Z}_p[T], \deg(r) < d,$$

such that

$$h(T) = q(T)g(T) + r(T).$$

This is the main tool that lets us isolate a polynomial factor from $F(T)$.

Step 4: construct the distinguished factor

We build sequences of approximations modulo powers of p .

We seek

$$F(T) \equiv g_r(T)w_r(T) \pmod{p^r R},$$

where:

- ▶ $g_r(T)$ is distinguished of degree d modulo p^r ,
- ▶ $w_r(T) \in R^\times$,
- ▶ and the pair (g_r, w_r) is compatible as r increases.

Starting from the factorization of $\overline{F}(T)$ in $\mathbb{F}_p[[T]]$, one lifts the factorization step-by-step from mod p^r to mod p^{r+1} .

Step 5: the lifting step

Assume we have already constructed g_r, w_r such that

$$F(T) \equiv g_r(T)w_r(T) \pmod{p^r R}.$$

Write

$$F(T) - g_r w_r = p^r E_r(T).$$

Using Weierstrass division by the distinguished polynomial $g_r(T)$, we write

$$E_r(T) = q_r(T)g_r(T) + s_r(T), \quad \deg(s_r) < d.$$

Now correct the factors by setting

$$g_{r+1}(T) = g_r(T) + p^r s_r(T), \quad w_{r+1}(T) = w_r(T) + p^r q_r(T).$$

Then a direct expansion shows

$$F(T) \equiv g_{r+1}(T)w_{r+1}(T) \pmod{p^{r+1}R}.$$

Step 6: pass to the limit

The sequences $\{g_r\}$ and $\{w_r\}$ are compatible modulo higher and higher powers of p .

Because R is complete for the (p, T) -adic topology, the limits exist:

$$g(T) = \lim_{r \rightarrow \infty} g_r(T), \quad w(T) = \lim_{r \rightarrow \infty} w_r(T).$$

Then

$$F(T) = g(T)w(T),$$

where: $g(T)$ is a distinguished polynomial, $w(T) \in R^\times$. Therefore

$$f(T) = p^m g(T)w(T).$$

Uniqueness follows from comparing the expression.